GPR Full-Waveform Inversion With Deep-Learning Forward Modeling: A Case Study From Non-Destructive Testing

Ourania Patsia, Antonios Giannopoulos, and Iraklis Giannakis

***Abstract*— Numerical modeling of ground penetrating radars (GPRs), such as the finite-difference time-domain (FDTD) method, has been extensively used to enhance the interpretation of GPR data and as a key component of full-waveform inversion (FWI). A major drawback of numerical solvers, especially within the context of FWI, is that they are still computationally expensive requiring often unattainable computational resources and access to high-performance computing (HPC). In this work, we present a near real-time deep-learning forward solver for GPR data that can generate entire B-scans, given certain model parameters as inputs. The machine-learning (ML) model is tuned for reinforced concrete slab scenarios, but the same rationale can be applied in a straightforward manner to other applications as well. The training was performed using entirely synthetic data, where a 3-D digital twin based on the 2000-MHz "palm" antenna from Geophysical Survey Systems, Inc. (GSSI) was included in FDTD simulations for the training set. The accuracy of the deep-learning solver is demonstrated with both synthetic and real data from reinforced concrete slabs. The predicted ML responses were in very good agreement with FDTD, showing a high degree of accuracy. The ML solver is then used as part of an FWI algorithm to characterize the concrete slab and estimate the depth and radius of the buried rebars. Coupled FWI with an ML-based forward solver results in significantly less execution times compared to conventional FWI using numerical solvers. The high accuracy of the proposed FWI, combined with the efficiency and speed of the ML-based forward solver, make the proposed scheme an ideal tool for characterizing concrete structures in nondestructive testing.**

***Index Terms*— Concrete, deep learning, finite-difference time-domain (FDTD), forward problem, full-waveform inversion (FWI), ground penetrating radar (GPR), machine learning (ML), neural networks (NNs).**

I. Introduction

GROUND penetrating radar (GPR) is a mainstream tool for concrete scanning and infrastructure assessment [1]. GPR has been widely used for locating structural elements in reinforced concrete slabs such as reinforcing bars (rebars), post-tension cables, and so on, locates voids and conduits, measures the slab thickness, and evaluates the condition and integrity of the concrete [2], [3], [4], [5]. In particular, when it comes to locating the reinforcement in concrete structures, GPR is very effective due to the high contrast between the dielectric properties of concrete and steel rebars. As with all targets, recovering quantitative information about rebars—such as their radius—from the GPR data represents a challenging task [6]. Hyperbola fitting is a well-known technique in GPR and has been applied in many cases to acquire estimates of the radius of rebars [7], [8], [9]. However, the accuracy of hyperbola fitting for estimating the radius of cylindrical objects is greatly compromised from nonuniqueness making the problem ill-posed and sensitive to noise [10]. Due to that, conventional GPR processing can only provide rough approximations when it comes to the radius unless the velocity of the medium is known [10]. In [11], a method for estimating the radius of a cylindrical target subject to velocity constraints is proposed and in [12], a method for more accurate velocity characterization taking into account the radius of the target was presented. A Kirchhoff linearized inversion scheme is successfully implemented in [13] to acquire rough information about the shape and size of metallic targets, while in [14], a Hough transform is employed to obtain estimates of buried pipe diameters. Assuming that the medium velocity is known, [15] uses a stationary wavelet transform to approximate the diameter of rebars. Full-waveform inversion (FWI) schemes can be used to simultaneously estimate the radius of cylindrical targets and the medium properties without the uncertainties and limitations of conventional GPR processing methods [16].

An essential component for implementing FWI is a forward solver. Electromagnetic (EM) forward solvers are frequently used to understand EM wave propagation and enhance the interpretation of GPR data [17], but also as part of FWI interpretations. FWI is a computationally intensive method that requires the execution of a forward problem multiple times and, therefore, a fast-forward solver can greatly increase the overall speed of FWI. The most common forward solver in GPR is the finite-difference time-domain (FDTD) method [18], which has been extensively used within the GPR community for simulating diverse and complex scenarios [19], [20], [21], [22]. Due to the advancements in computational resources over the last years, a full 3-D FDTD solver is now possible, but still remains a computationally expensive algorithm and often unattainable in the absence of

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Ourania Patsia and Antonios Giannopoulos are with the School of Engineering, Institute for Infrastructure and Environment, The University of Edinburgh, EH9 3FG Edinburgh, U.K. (e-mail: O.Patsia@ed.ac.uk; A.Giannopoulos@ed.ac.uk).

Iraklis Giannakis is with the School of Geosciences, University of Aberdeen, AB24 3FX Aberdeen, U.K. (e-mail: iraklis.giannakis@abdn.ac.uk).

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high-performance computing (HPC), especially when used as part of an FWI scheme. Therefore, although FWI is proven capable of estimating the radii or rebars, its applicability, in practice, is limited due to the lack of a fast-forward solver. The current article describes a forward solver based on machine learning (ML) that can simulate whole B-scans in near real time, given certain model parameters as inputs.

ML is increasingly used in the last years for different GPR processing and interpretation applications. In [23], [24], and [25], artificial neural networks (NNs) are implemented for buried object detection in B-scan images, whereas [26] uses transfer learning for target classification. Fully connected feed-forward (Dense) NNs are utilized by [27] for landmine detection and classification, while in [28], a hybrid NN structure with both supervised and unsupervised learning is used to detect and classify different types of landmines. For the same task, a different approach using convolutional NNs (CNNs) for B-scan image classification was proposed in [29] and [30]. A clutter suppression algorithm based on autoencoders (AEs) was presented in [31]. In [32], NNs and random forest (RF) are suggested for estimating the diameter of rebars in concrete slabs, while in [33], deep learning was employed to estimate both the geometrical characteristics and the dielectric properties of buried cylinders.

The first development of a fast ML-based forward solver for GPR data from concrete slabs was described in [6] and [34]. The scheme utilizes a dense network along with principal component analysis (PCA) to predict A-scan responses for which the target is buried exactly below the center of the antenna. This forward solver was used as part of FWI to estimate the water content of concrete and the depth and diameter of rebars. The scheme included a model of a 1500-MHz transducer from Geophysical Survey Systems, Inc. (GSSI) [35], [36]. FWI using the ML model as a forward solver was validated with both synthetic and real data demonstrating the accuracy of the scheme. The main drawback of the ML-based forward solver presented in [6] and [34] is the fact that it can only simulate single traces in scenarios where the rebar is placed exactly underneath the antenna. This constrains the applicability of the technique since the ML-based forward solver cannot simulate whole B-scans. This limitation is addressed in the current article by building upon [6], [34] and developing a novel ML-based solver capable of simulating whole B-scans in near real time.

In particular, a deep-learning forward solver is presented, which is used as part of an FWI algorithm to characterize concrete and estimate the depth and diameter of rebars. The scheme presented here follows the work by [6] and was extended to make predictions for different rebar lateral positions relative to the antenna center. Therefore, entire B-scans can be generated fast in almost real time reducing significantly the computational requirements of FWI. The early stages of the work and the primary network architecture were developed in [37], while the network has been further optimized in the present work to tune the hyperparameters for the specific problem and improve the overall performance. The proposed ML scheme is analytically described here and is applied to additional examples, verifying the effectiveness of the method. The deep-learning model in conjunction with FWI was validated using both synthetic and real data, supporting the premise that FWI coupled with ML-based forward modeling can be a powerful, practical, and commercially appealing methodology for concrete inspection using GPR.

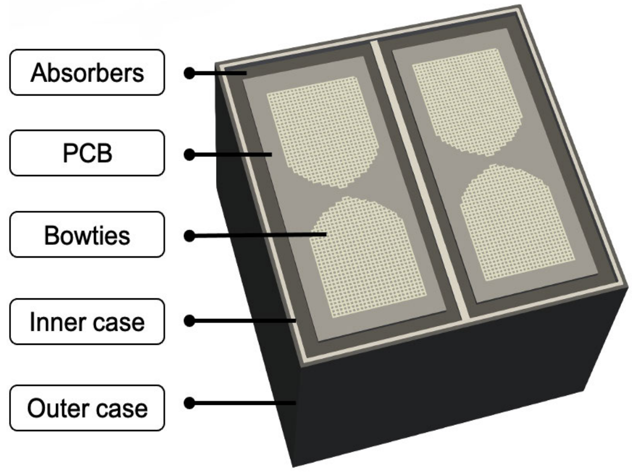

Fig. 1. Modeled geometry of the GSSI 2000-MHz "palm" antenna.

II. Deep-Learning Forward Solver

The current article suggests an FWI scheme for characterizing concrete structures and estimating the depth and radius of rebars. The proposed FWI is coupled with a novel deep-learning-based forward solver that greatly accelerates the performance of FWI, resulting in an accurate and at the same time practical GPR tool. This section describes the training process and the architecture of the deep-learning forward solver, focusing on the generation of the synthetic training data—the most crucial step in the whole process.

A. Training Set

To accurately map the causal relationship between given inputs and their corresponding labels using supervised ML, a sufficient amount of equally distributed training data is needed [38]. For concrete inspection using GPR, this would require the acquisition of thousands of data in well-controlled environments with known targets and concrete properties. Such an experimental setup is unattainable primarily due to the lack of ground truth, necessary in supervised learning. To overcome this, and similar to [6], the training data for the proposed ML-forward solver scheme consists entirely of synthetic data, which were generated using gprMax [39], an open-source FDTD solver. The domain size is set to 300 $\times$ 300 $\times$ 400 mm with a 1-mm step size and a time window of 8 ns, resulting in a total of 4156 time steps.

A numerical equivalent of the GSSI 2000-MHz "palm" antenna was used for the generation of the training data. The development of the digital twin and details regarding its structure can be found in [40]. The FDTD geometry of the model is illustrated in Fig. 1. The model comprises two planar surface bowties, with additional rectangular extensions at each end of the bowtie arms. The bowties are laid on top of printed

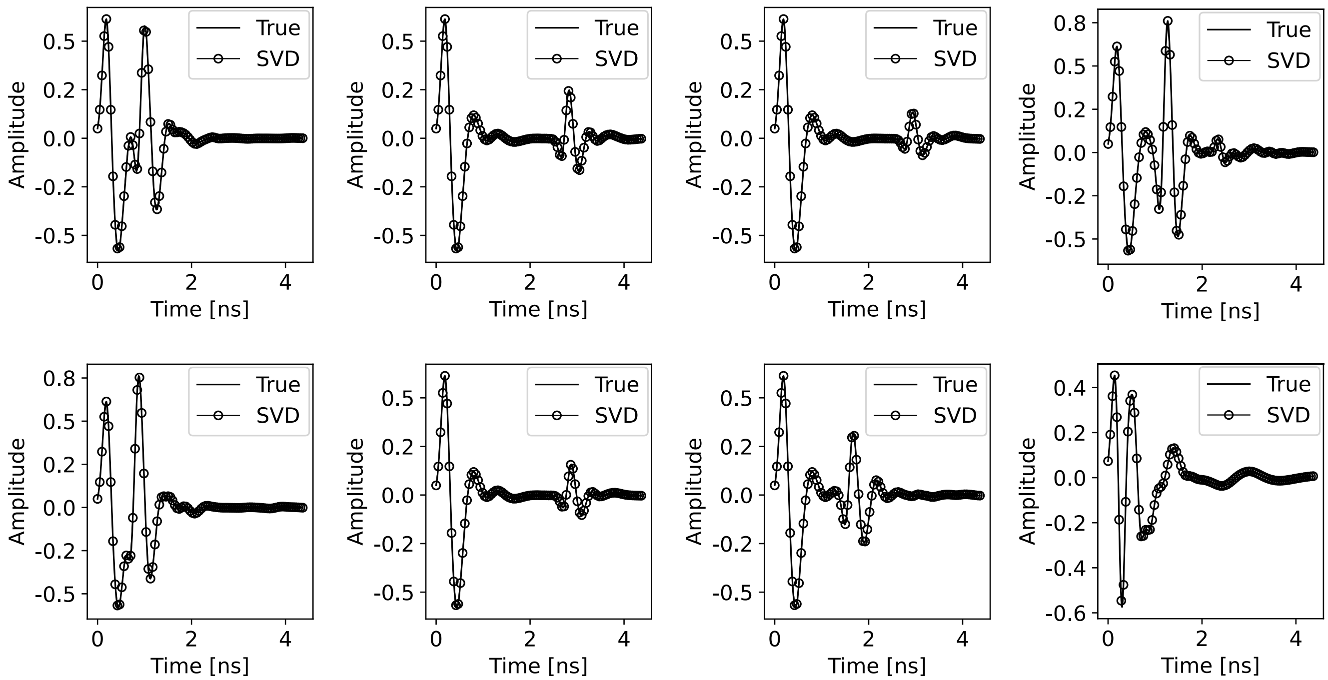

Fig. 2. Compressed A-scans using 70 SVD components, plotted versus true A-scans randomly selected from the training set.

TABLE I
DEBYE PROPERTIES OF CONCRETE

MC (%)	ϵ_s	ϵ_∞	t_0 (ns)	σ_{DC} (S/m)
12	12.84	7.42	0.611	20.6×10^{-3}
9.3	11.19	7.20	0.73	23×10^{-3}
6.2	9.14	5.93	0.80	6.7×10^{-3}
5.5	8.63	6.02	1.00	5.15×10^{-3}
2.8	6.75	5.50	2.28	2.03×10^{-3}
0.2	4.8	4.50	0.82	6.06×10^{-4}

circuit boards (PCBs). Underneath each PCB, there are two different layers of EM absorber foams, which are utilized to reduce the back-cavity radiation. Both antennas are situated in the same inner metallic case and are further enclosed in a plastic outer case. A voltage source is used to excite the model along with a Gaussian-shaped pulse with a 2.12-GHz central frequency.

The antenna is coupled to a homogeneous and dispersive concrete medium that is modeled using an extended Debye model [41]

$$\epsilon = \epsilon_\infty + \frac{\epsilon_s - \epsilon_\infty}{1 + j\omega t_0} + \frac{\sigma}{j\omega\epsilon_0} \tag{1}$$

where ϵ_0 is the absolute permittivity of vacuum, ϵ_∞ is the relative permittivity at infinite frequency, ϵ_s is the relative permittivity at zero frequency, t_0 is the relaxation time, σ is the conductivity, and ω is the angular frequency. The values of the extended Debye model parameters for concrete are provided in Table I and were measured experimentally in [41]. From Table I, it is evident that the Debye properties of concrete are related just to moisture content, reducing the number of parameters from 4 to 1. Via this approach, we avoid instabilities that might arise in FWI due to the sensitivity differences between extended Debye parameters.

The rebars are modeled as perfect electric conductors (PECs), and the main axis of the antenna (E-field) was parallel to the main axis of the rebars in all of the simulations. This implies that the proposed scheme is applicable only to B-scans perpendicular to the main axis of the rebars, a typical measurement configuration necessary to many GPR processing tools such as hyperbola fitting.

To acquire a set of diverse scenarios, cubic interpolation was used to interpolate between the values shown in Table I and obtain Debye properties that correspond to different moisture content between 0.2% and 10%. Using these properties as input to simulate concrete in the FDTD algorithm, 4000 traces of different concrete slab scenarios were generated and used to train the ML model. For each model, the water fraction (MC), the diameter (R), depth (D), and position (X) of the rebar relative to the antenna center are randomly drawn from a uniform distribution over the ranges.

1) $R \in [1, 6]$ cm.
2) $D \in [3, 25]$ cm.
3) $X \in [-11, 11]$ cm.
4) $MC \in [0.2 - 10]$%.

The four parameters, R, D, X, and MC, are used as inputs to the ML-based solver for simulating in near real-time the resulting A-scan (output). The training data are subjected to preprocessing to transform them into a form that is more suitable for training. Typical normalization was applied both to inputs and outputs, and the A-scans were downsampled to 512 time steps to decrease the training time.

Finally, since the A-scan data are in a high-dimensional space, it is essential to compress them and acquire a

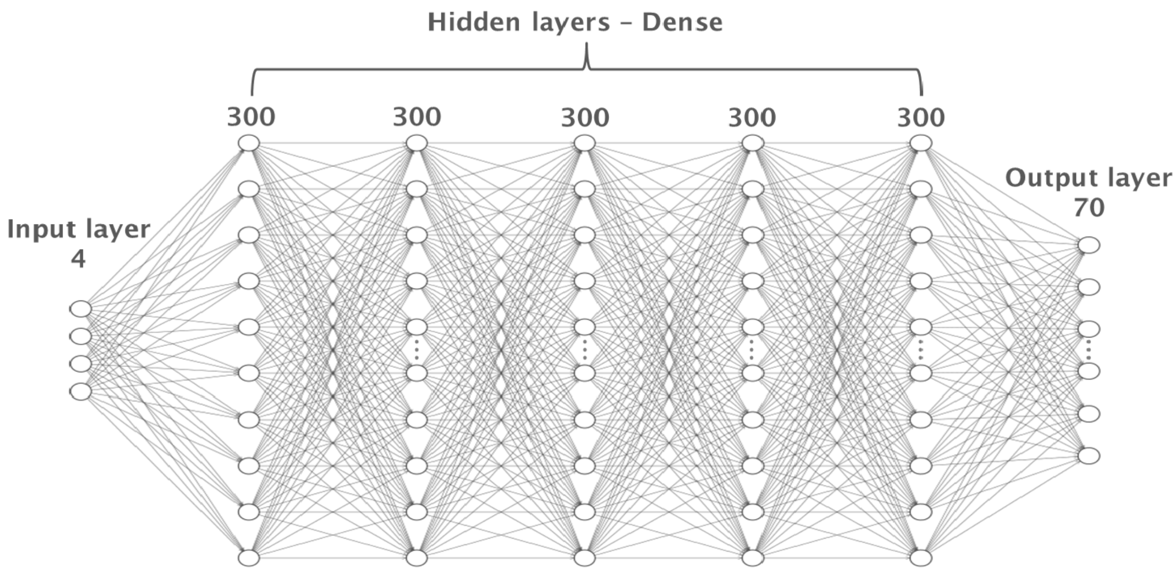

Fig. 3. NN architecture of the proposed ML forward solver. The network consists of five dense hidden layers with 300 nodes each, an input layer with four nodes, and an output layer with 70 nodes.

lower-dimensional representation, which will enhance the performance of the ML algorithm. To compress the data, singular value decomposition (SVD) was implemented. SVD has been used by the GPR community for direct wave removal and other clutter reduction [42], [43], [44], [45], while in ML is commonly used as a data compression method. 70 SVD components were found to be sufficient to capture the variability in the data. Conclusively, the four inputs R, D, X, MC are used to estimate the compressed representation of the corresponding A-scan (just 70 values), which is subsequently decompressed using SVD to generate the time domain A-scan. To demonstrate the effectiveness of SVD on compressing A-scans, a number of SVD reconstructed A-scans from the training set is plotted versus the true responses in Fig. 2. The actual traces are almost identical to the compressed ones indicating the effectiveness of the SVD compression with no significant loss of information.

Once the data have been generated and processed, they are shuffled and randomly split into training, test, and validation set. An 80% of the data is used as the training set, whereas the validation and test sets are allocated a 10% each. The training set is used to train the deep-learning model, the validation set is used to evaluate the performance of the model during the process of training and fine-tune it, while the test set is used to assess the performance of the final model on previously unseen data.

B. Deep-Learning Scheme

The ML-based forward solver is a feed-forward fully connected (Dense) type of network that consists of five hidden layers with 300 nodes at each layer. The input layer includes four nodes, since there are only four input parameters (R, D, X, MC), while the output layer consists of 70 nodes representing the SVD compressed version of each A-scan. Fig. 3 illustrates the architecture of the network. The rectified

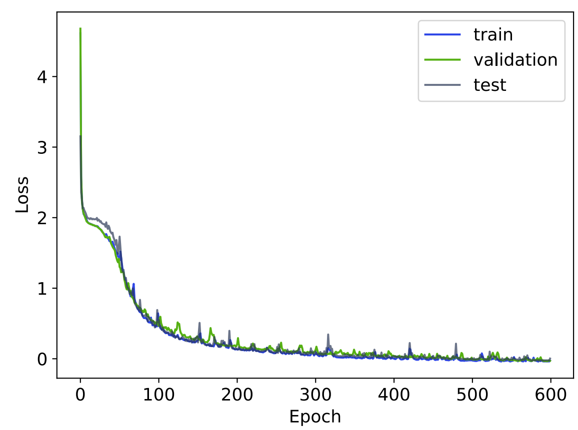

Fig. 4. Training, validation, and test set loss values per epoch.

linear unit (ReLU) is used as an activation function for all layers, except the output layer, where a linear activation function is used. The loss function chosen to be minimized with the Adam optimizer is the mean squared error (mse). The learning rate is set to 0.001 with a 1e-7 decay at each iteration. Ensemble averaging was used, where 40 different ML models were developed. All models have the same architecture but the difference is that each ML model is trained with different initialization weights. Each ML model is trained separately from the rest of the models, the output is predicted from each model, and all the outputs are averaged out to provide the final result. Using multiple models helped the algorithm to generalize and increased accuracy.

C. Results

The final ML model resulted in an mse error of 4.64e-5 for the training set, 9.88e-5 for the validation set, and 6.1e-4 for the test set. The error convergence curves per epoch for the three sets are shown in Fig. 4. All the curves converge

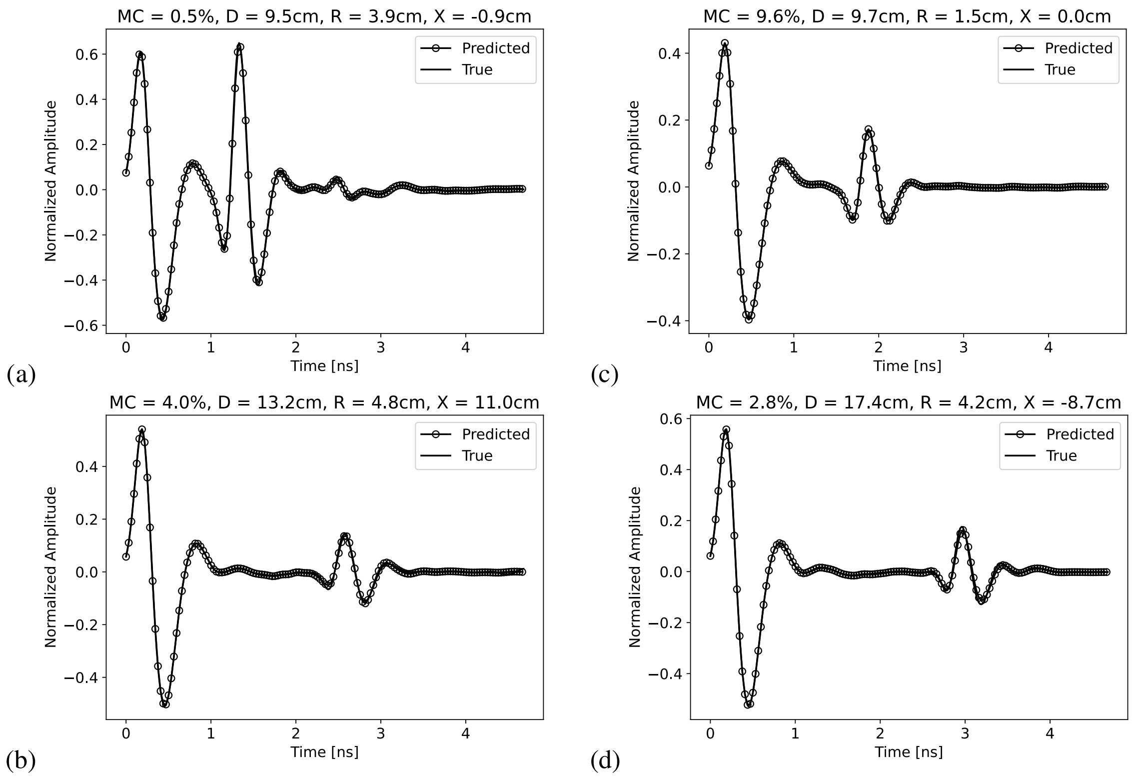

Fig. 5. Comparison between the FDTD and the ML-based forward solver in four different case studies. The properties used for each case are shown at the top of each subfigure. (a) Test case 1, (b) Test case 2, (c) Test case 3, and (d) Test case 4.

similarly, starting from a high loss in the first iterations toward a low loss at the last iterations with the flattening of the curves showing the convergence. The low testing errors demonstrate that the model has not overfitted the training set and can generalize to new unseen cases.

Fig. 5 illustrates a comparison between A-scans from FDTD and the ML-based forward solver in four unknown cases that were not included in the training set. The model parameters used in each case are shown in the figure. In all cases, the ML predictions follow closely the pattern of the ground truth and are almost indistinguishable from the FDTD traces. It is obvious that the ML forward solver predicts successfully both the waveform shape and the arrivals of the responses for different water content, depth, and diameter of rebars, even in cases where the target is not buried directly below the transducer, demonstrating its ability to generate entire B-scans. To further demonstrate this, B-scans from two scenarios were generated using both FDTD and the ML solver and are presented in Fig. 6. The model parameters that each B-scan corresponds to are again shown in the figure. The B-scans generated by the ML model are almost identical to the ones generated using FDTD with an average cross correlation between A-scans 99.2% for the first case and 99.5% for the second case.

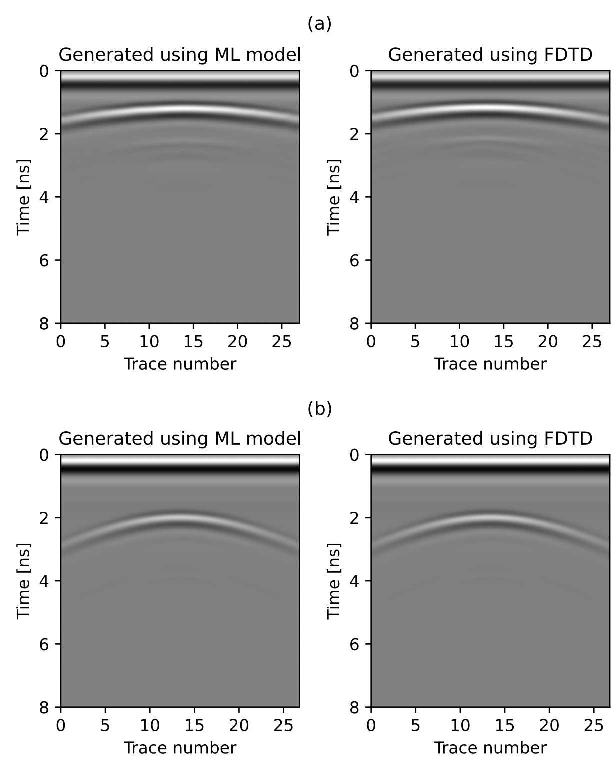

Fig. 6. Comparison between the FDTD and the ML model B-scans for two different cases. The model properties used for each case are shown at the top of each row of subfigures. (a) $MC = 3.9\%$, $D = 7.0$ cm, and $R = 2.5$ cm. (b) $MC = 5.1\%$, $D = 12.0$ cm, and $R = 2.0$ cm.

III. Full Waveform Inversion

The ML-based forward solver is now coupled with an FWI scheme to simultaneously estimate R, D, X, MC. The algorithm is validated using both synthetic and real data to assess its performance and map its limitations.

A. Numerical Case Studies

A synthetic B-scan was generated using FDTD for a generic scenario where a rebar is placed inside a concrete medium. The

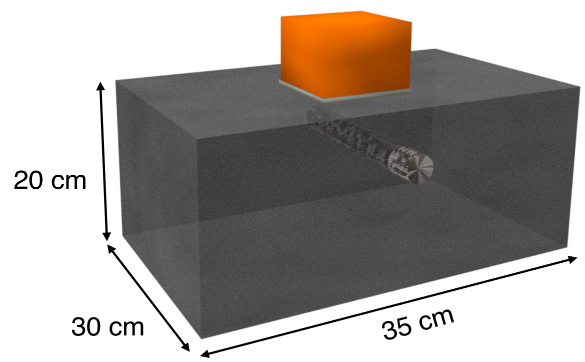

Fig. 7. Investigated scenario for testing FWI coupled with the novel ML-based forward solver. Metallic rebar with $R = 3$ cm is buried at $D = 7.8$ cm, while concrete was simulated with a water content of $MC = 3.5\%$. The antenna is a numerical equivalent of the GSSI 2000-MHz "palm" antenna.

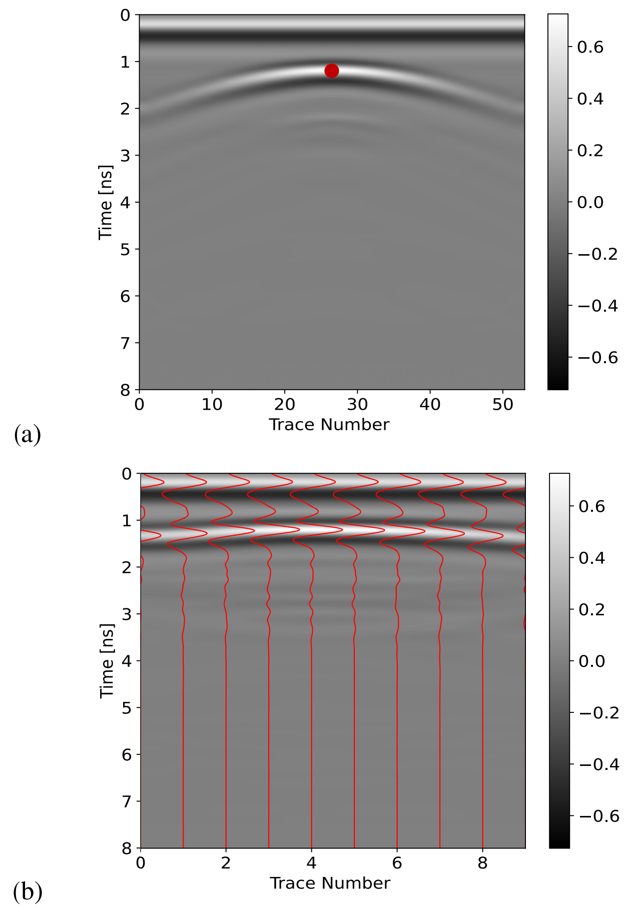

Fig. 8. (a) Resulting B-scan for the synthetic scenario shown in Fig. 7. (b) Ten traces selected to be used in FWI.

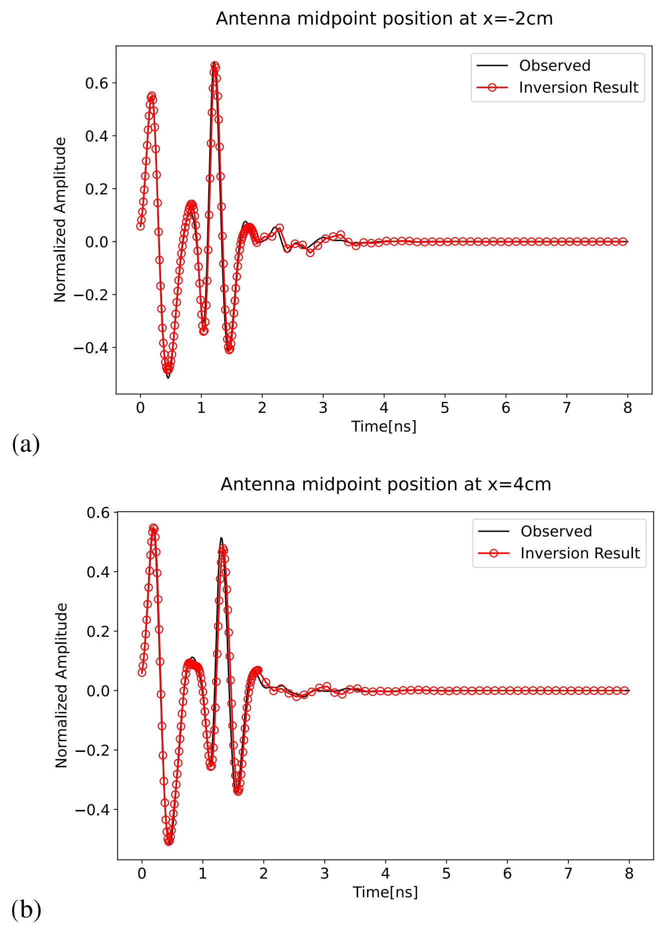

Fig. 9. Comparison of FWI results with observed data for the numerical case study shown in Fig. 7. (a) Antenna center positioned 2 cm on the left of the target and (b) antenna center positioned 4 cm on the right of the target.

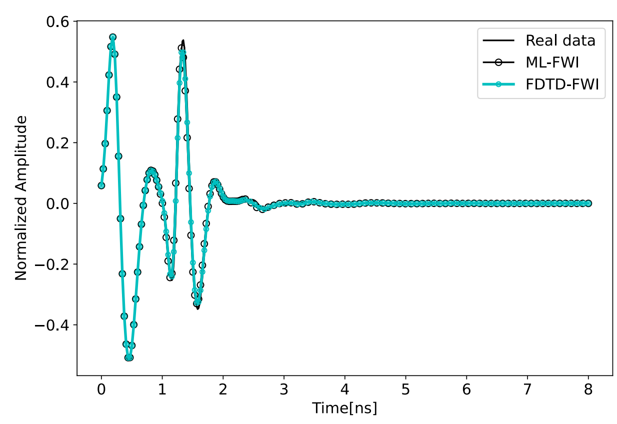

Fig. 10. Comparison of ground-truth data with the ML-FWI and FDTD-FWI results for the numerical case study shown in Fig. 7 and for rebar position $x = 3$ cm.

model consists of one rebar with diameter $R = 3$ cm buried at a depth of $D = 7.8$ cm, while concrete was simulated with a water content of $MC = 3.5\%$. The geometry of the model is shown in Fig. 7 and the resulting B-scan is illustrated in Fig. 8(a). A total of ten traces with 1 cm spacing were used in FWI. The traces were chosen relative to the apex of each hyperbola, as demonstrated in Fig. 8(b).

To perform the inversion, a global optimizer, the particle swarm optimization (PSO) was implemented. The loss function chosen to be minimized is the mean absolute error between the synthetic traces and the A-scans resulting from the inversion. Although a global optimizer is a computationally expensive algorithm, it was chosen to overcome the problem of local minima, which is encountered in gradient-based FWI algorithms. The use of global optimizers is attainable since the ML-based forward solver is nearly real time and

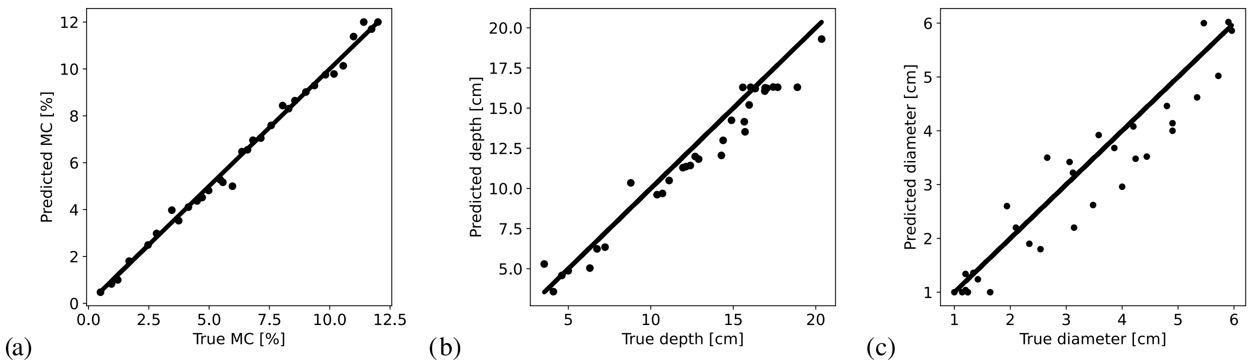

Fig. 11. Comparison between true with the predicted parameters from FWI for 30 different numerical case studies. (a) Water fraction (MC). (b) Depth of rebar (D). (c) Diameter of rebar (R).

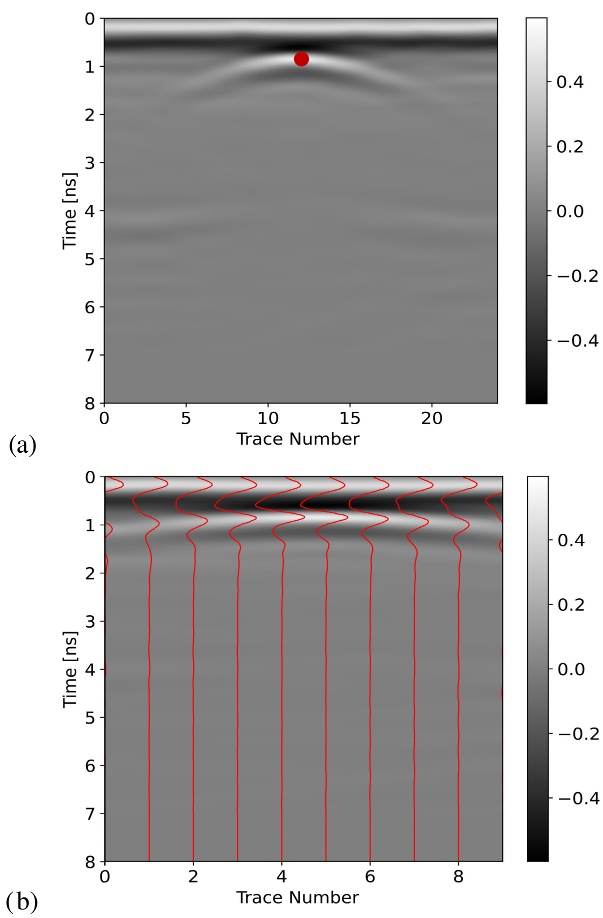

Fig. 12. (a) Data collected from the real case study shown in Fig. 13. (b) Ten traces selected to be used in FWI.

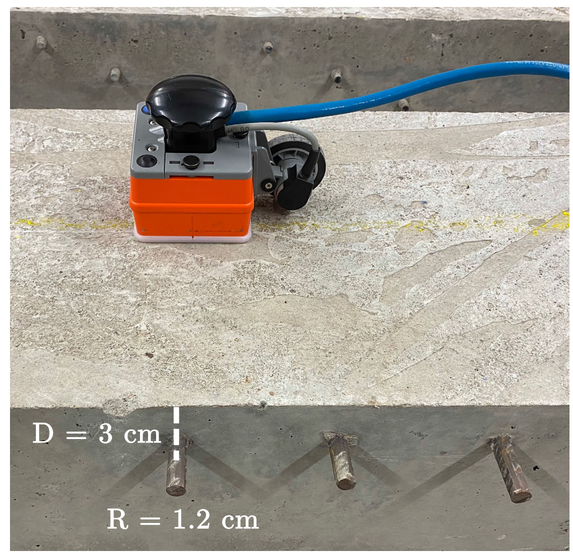

Fig. 13. Setup of the concrete slab scenario used along with the GSSI 2000-MHz transducer. The measurements were taken perpendicular to the main axis of a metallic rebar with $R = 1.2$ cm buried at $D = 3$ cm.

TABLE II

FWI RESULTS FOR THE NUMERICAL CASE STUDY SHOWN IN FIG. 7

	MC(%)	Depth (cm)	Diameter (cm)
Ground truth	3.5	7.8	3
ML-FWI	3.44	7.4	2.95
FDTD-FWI	3.52	7.2	2.88

TABLE III

FWI RESULTS FOR THE REAL CASE STUDY SHOWN IN FIG. 13

	MC(%)	Depth (cm)	Diameter (cm)
Ground truth	-	3	1.2
ML-FWI	11.2	3.6	1.26
FDTD-FWI	10.9	3.8	1.22
[6]	11.5	-	-

has minimum computational requirements. Global optimizers using conventional FWI coupled with FDTD are computationally unattainable with access to HPC.

The actual and the estimated parameters from FWI are given in Table II. It is apparent that the obtained values are very close to the true parameters. Fig. 9 displays the observed A-scans for which the target is located 2 cm on right and 4 cm on the left of the antenna center, compared with the responses produced using the FWI resultant parameters. The traces are almost identical, validating the accuracy of the solution.

To get a sense of how similar the ML-based forward solver behaves to FDTD as part of FWI, the same FWI scheme was applied to the synthetic data using gprMax as the FDTD solver. The estimated parameters from the FDTD-FWI, which are also given in Table II, are very close to the ML-FWI predicted values and also to the true ones. To further evaluate the accuracy and similarity of the schemes, the FWI traces

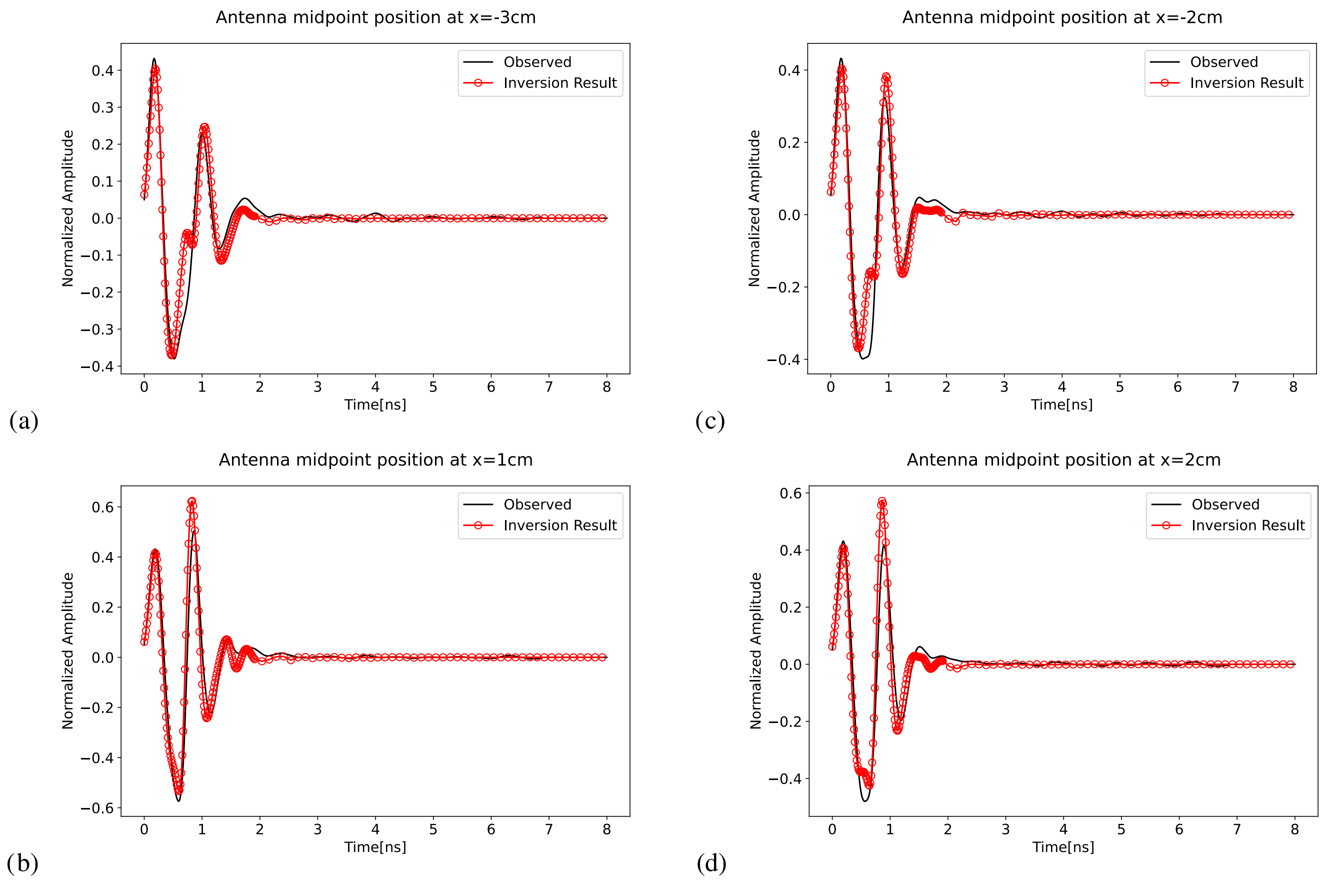

Fig. 14. Comparison between the real and inverted A-scans using the proposed FWI for the real case study shown in Fig. 13. (a) Antenna center positioned 3 cm on the left of the target, (b) antenna center positioned 1 cm on the right of the target, (c) antenna center positioned 2 cm on the left of the target, and (d) antenna center positioned 2 cm on the right of the target.

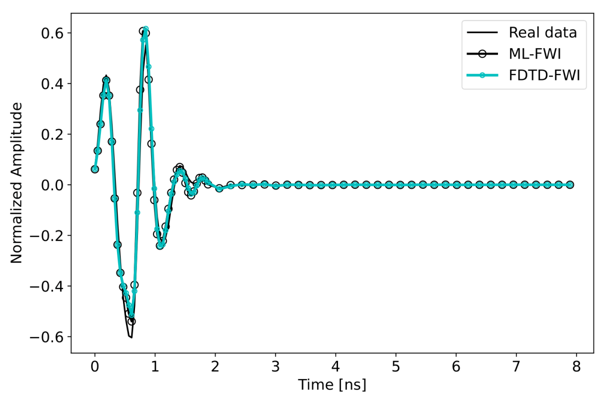

Fig. 15. Comparison of real data with the ML-FWI and FDTD-FWI results for the real case study shown in Fig. 13 and for rebar position $x = 1$ cm.

were investigated. Fig. 10 shows a comparison between the actual synthetic A-scan, the ML-FWI, and the FDTD-FWI traces for rebar position at $x = 3$ cm, where the three responses are almost identical. The above show that the proposed ML forward solver results in similar accuracy to FDTD.

The same ML-FWI scheme was applied to 30 different randomly selected case studies with different R, D, X, MC. The predicted FWI parameters along with the true parameters are shown in Fig. 11. It is obvious that the three parameters can be accurately predicted within a reasonable margin of error. The water content predictions were the most accurate, predicting values very close to the true ones in all cases. The depth and diameter follow with an error of ± 1 and ± 0.5 cm, respectively, which are acceptable values for industry standards. Note that although ten traces were used in the inversion, the scheme can provide predictions with similar accuracy with even less selected traces. The FDTD-FWI scheme was also applied to the same 30 case studies, yielding similar predicted values to the ML-FWI algorithm with a maximum difference between the two schemes $\pm 0.13\%$ for MC, ± 0.6 cm for depth, and ± 0.6 cm for radius.

B. Real Data

Since a digital twin of a real transducer, which replicates the behavior of the real antenna, was used in the simulations, the ML-FWI algorithm can be applied to real GPR data collected with the GSSI 2000-MHz "palm" antenna. Real data from reinforced concrete slabs were measured in the NDT Laboratory at the University of Edinburgh in the School of Engineering. Fig. 12 displays one of the B-scans gathered over a single rebar of a concrete slab, while the concrete slab along with the transducer is shown in Fig. 13. The same FWI scheme, that was applied to the synthetic data, was also applied to the real case study.

The estimated rebar depth and diameter are very close to the ground truth as shown in Table III. Although the water content of the concrete was unknown and could not be easily determined, the value of 11.2% that was obtained agrees with the value of 11.5% obtained in [6] for the same concrete batch and using a different GSSI transducer of 1.5 GHz. The resulting traces, obtained using the FWI predicted parameters, compared with the real responses are shown in Fig. 14 for different antenna positions, showing a good agreement in all cases.

Similar to the synthetic data, a comparison was made between the real data, the ML-FWI, and the FDTD-FWI scheme. The estimated values from FDTD-FWI for the water content, depth, and diameter of rebar were 10.9%, 3.8, and 1.22 cm, respectively, which are in very good agreement with both the ground truth and the values obtained using the ML solver, as displayed in Table III. Fig. 15 shows a comparison between the real A-scan, the ML-FWI, and the FDTD-FWI traces for rebar position at $x = 1$ cm, showing a very good similarity between the three. This further supports the premise that the ML-based forward solver is similar to the FDTD solver and at the same time orders of magnitude faster.

IV. Execution Times

Generating the entire training set of 4000 traces required a run time of ~2 days on an NVIDIA TITAN RTX 24 GB GPU given that a single A-scan generated using FDTD takes 40 s, utilizing the GPU capabilities of gprMax [46]. The ML forward solver can predict an A-scan instantly. The FWI algorithm with an FDTD solver and using ten traces, 40 iterations, and 40 particles required a runtime of ~8.5 h on 4 NVIDIA TITAN RTX 24 GB GPUs, which would have been increased to ~35 h if a single GPU was used. In contrast, the FWI-ML scheme using the same inversion options as the FWI-FDTD required a runtime of 1.5 h on an Intel[1] Xeon[1] Silver 4210 CPU @ 2.20 GHz without utilizing parallelization which would have decreased even further the execution time.

V. Conclusion

In this article, we presented a deep-learning-based forward solver that can accurately predict in real-time A-scans even when the antenna is not placed directly above the target, thus producing entire B-scans. The solver is adapted for reinforced concrete slab scenarios but can be tuned for other cases. Training of the ML model was performed using synthetic FDTD data that represented realistic scenarios and included a digital twin of a real transducer. The deep-learning model was used as part of FWI to acquire estimates of the concrete water content, depth, and diameter of rebars. Applying the ML-FWI scheme to a large number of both synthetic and real data proved its accuracy in acquiring estimates of key properties, validating its performance. Comparison of the ML-solver with FDTD showed that it can accurately replicate the FDTD responses in orders of magnitude less computational time. Having an accurate and almost real time forward solver, makes FWI, even when implemented using global optimizers, applicable to realistic scenarios in practice.

[1]Registered trademark.

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Ourania Patsia received the B.Sc. degree in geology and the M.Sc. degree in geophysics from the Aristotle University of Thessaloniki, Thessaloniki, Greece, in 2017 and 2021, respectively, and the Ph.D. degree from the School of Engineering, Infrastructure and Environment, The University of Edinburgh, Edinburgh, U.K., in 2022.

She is currently a University Teacher with the School of Engineering, The University of Edinburgh. Her research interests include near-surface geophysics, computational electromagnetics, and machine learning. She is working on machine-learning techniques applied to ground-penetrating radar problems.

Antonios Giannopoulos received the B.Sc. degree in geology from the Aristotle University of Thessaloniki, Thessaloniki, Greece, in 1991, and the D.Phil. degree in electronics from The University of York, York, U.K., in 1997.

He is currently a Reader and the Director of the civil and environmental engineering discipline with the School of Engineering, The University of Edinburgh, Edinburgh, U.K. He created gprMax, a freely available FDTD simulator, and he is directing its continuous development and enhancement. His research interests include computational electromagnetics and in particular the application of the FDTD method on the numerical modeling of ground-penetrating radar. In addition, he works on the development and application of advanced radar and geophysical techniques for infrastructure sensing applications.

Dr. Giannopoulos was the General Chair of the 9th International Workshop on Advanced Ground Penetrating Radar, Edinburgh, 2017. He is a member of SEG and EAGE.

Iraklis Giannakis received the bachelor’s and master’s degrees in geophysics from the Aristotle University of Thessaloniki, Thessaloniki, Greece, in 2009 and 2011, respectively, and the Ph.D. degree from The University of Edinburgh, Edinburgh, U.K. in 2016.

He was co-funded by the Defense Science and Technology Laboratory, Salisbury, U.K., and the Engineering and Physical Sciences Research Council, Swindon, U.K., through the Ph.D. project. He is currently a Lecturer with the School of Geosciences, University of Aberdeen, Aberdeen, U.K. His research interests include near-surface geophysics, nondestructive testing, computational electromagnetics, and machine learning.